SWE 3513 / INDIVIDUAL ASSIGNMENT 2

01. Project identity and decision context

Musanze Harvest Intelligence Service

IDENTIFICATION	DETAIL
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Course	SWE 3513 — Artificial Intelligence

Problem definition

This project develops a classroom decision-support prototype for a potato cooperative field officer in Musanze. The system answers three connected questions: how much harvest a farm may produce, whether its production conditions indicate high risk, and which farms share similar conditions.

Decision-support scope

The solution combines a numerical harvest estimate (regression), a binary high-risk prediction (classification), and descriptive grouping of farm conditions (clustering). Users can request predictions from a command-line interface or a local FastAPI service.

Ethical boundary

The data are synthetic and the predictions are educational demonstrations. Outputs must not replace agronomist judgement or independently determine eligibility for finance, insurance, or public services.

02. Dataset and reproducibility

A controlled synthetic experiment, not field-collected evidence

MEASURE	OBSERVED VALUE
Personal random seed	7123
Base records	270 (240 + 7123 mod 41)
Generated CSV records	273, including 3 exact duplicates
Cleaned records	270 after duplicate removal
Intentional missing values	14 soil moisture; 14 fertilizer

Input features and targets

INPUT FEATURE	RANGE / VALUES
rainfall_mm	600–1400 mm
avg_temp_c	12–26 °C
soil_moisture_pct	20–80%
fertilizer_kg	0–250 kg
field_size_ha	0.2–4.0 ha
season	A or B
cooperative_support	0 or 1

The regression target, yield_tons, is derived from the prescribed synthetic generation formula. The binary target, high_risk, is defined using below-median yield per hectare. The generator uses NumPy default_rng(7123) and writes numerical CSV fields to four decimal places.

SHA-256 fingerprint: 759d98eee29266a8d54d7de3c50c4a24c4ea50d4d5b38bf7082ed4ba3cb3e014

03. Experimental pipeline

Preprocessing, evaluation design, and leakage safeguards

Data checks and preprocessing

The input CSV is checked for schema and valid value ranges before cleaning. Exact duplicate rows are then removed. Numerical features use median imputation followed by standard scaling. The categorical season variable uses most-frequent imputation and one-hot encoding.

Split and validation

A 75/25 training/test split uses random_state=7123 and stratifies the classification target. Candidate supervised models are compared using five-fold cross-validation on the training partition. The held-out test data are reserved for final evaluation.

Preventing leakage

Preprocessing is encapsulated in scikit-learn ColumnTransformer and Pipeline objects so the transformations are fitted to training data rather than to the full dataset. The classifier never receives yield_tons, yield per hectare, or the high_risk target as predictors.

Models investigated

TASK	CANDIDATE APPROACHES
Regression	Mean baseline; linear regression; random forest
Classification	Majority baseline; logistic regression; random forest
Clustering	KMeans, candidate K from 2 through 6

Saved artifacts and inference

The fitted pipelines and supporting metadata are saved with joblib. They are subsequently loaded by the local command-line prediction tool and FastAPI service, avoiding the need to retrain for each request.

04. Supervised learning results

Candidate selection by training cross-validation; final reporting on held-out data

Regression: harvest quantity

Linear regression achieved the lowest five-fold training cross-validation MAE, 0.8380, and was selected for final testing. The comparison MAEs were 2.5205 for the mean baseline and 1.2587 for random forest.

HELD-OUT REGRESSION METRIC	VALUE
Mean absolute error (MAE)	0.7979 tons
Root mean squared error (RMSE)	1.0673 tons
Coefficient of determination (R^2)	0.8835

The saved residual plot (artifacts/plots/residuals.png) illustrates individual prediction errors. Since noise was added during synthetic generation, these residuals are not errors measured in operational field conditions.

Classification: production risk

Logistic regression was selected using the highest training cross-validation F1 score, 0.9396. Held-out evaluation produced the following results.

METRIC	VALUE
Precision	0.9375
Recall	0.8824
F1 score	0.9091
ROC-AUC	0.9862

The confusion matrix is [[32, 2], [4, 30]]: 32 true negatives, 2 false positives, 4 false negatives and 30 true positives. Four actual high-risk cases were missed. This underlines the importance of manual field-officer review.

05. Unsupervised grouping

Interpreting KMeans clusters with appropriate caution

Choosing the number of clusters

KMeans was assessed for K = 2 to 6 without using either target variable. The highest observed silhouette score was 0.1572 at K = 2, so the final model uses two clusters.

NUMBER OF CLUSTERS (K)	SILHOUETTE SCORE
2 (selected)	0.1572
3	0.1396
4	0.1411
5	0.1391
6	0.1493

Cluster profile

PROFILE	CLUSTER 0	CLUSTER 1
Mean cooperative support	1.0	0.0
Mean fertilizer quantity	138.1 kg	127.3 kg

Interpretation

The silhouette score of approximately 0.16 indicates weak separation. The two groups can support exploratory discussion about observed farm characteristics, but should not be treated as strong natural segments or evidence of causal differences.

The project saves a clustering plot at artifacts/plots/clusters.png. Because this is a descriptive unsupervised exercise, target fields were excluded from cluster fitting.

06. Delivery, testing and reflection

Operational checks and responsible future improvement

Prediction interfaces

The CLI loads the saved pipelines and returns harvest estimate, risk category, risk probability, cluster and verification code. The FastAPI service exposes GET /health, GET /model-info and POST /predict. Pydantic validates required input fields and ranges; invalid API requests return HTTP 422.

VALID SAMPLE RESPONSE FIELD	OBSERVED OUTPUT
Estimated harvest	12.2258 tons
Risk class / probability	0 / 0.2867
Cluster	0
Verification code	AI-A2-7123

An invalid CLI sample missing cooperative_support was rejected. Manual checks confirmed that GET /health and GET /model-info respond correctly, Swagger documentation loads, and POST /predict returns a valid prediction while a missing required field produces HTTP 422.

Software verification

The full pipeline reported completion, and the automated pytest suite returned 8 passed, 1 warning on the latest reported run (27.52 seconds). The warning concerned Starlette TestClient/httpx deprecation, rather than a failing test. The test suite covers identity and seed checks, reproducibility, data handling, validation, saved-model loading, CLI behavior and API behavior.

Limitations and next steps

The sample is small and entirely synthetic; its targets follow a generation formula rather than observed harvest outcomes. The clusters are weakly separated, and there has been no external validation. A future field study should test against actual harvest records, add soil and disease measurements, examine fairness and cooperative access effects, and monitor changing performance across growing seasons.

Final judgement remains with a qualified agronomist or cooperative field officer, who should verify measurements and review model outputs before taking any action.